

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) / S4 (PT) (S, FE) Examination December 2024 (2019 Scheme)

Course Code: EET306**Course Name: POWER ELECTRONICS**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | Explain the working principle of wide band gap devices | (3) |
| 2 | Explain the necessity of isolation in gate drive circuits | (3) |
| 3 | Draw the input and output voltage waveforms of a single phase fully controlled rectifier for $\alpha > 90^\circ$ in discontinuous conduction mode feeding to RL Load. | (3) |
| 4 | Draw the output voltage waveform of a 3-phase controlled half wave rectifier for $\alpha = 150^\circ$. | (3) |
| 5 | Derive the expression for output voltage and input power factor of a single phase full wave AC voltage controller with R load | (3) |
| 6 | Explain the working of current source inverter with a relevant diagrams | (3) |
| 7 | With a neat circuit diagram and waveforms explain the working of a buck converter | (3) |
| 8 | Describe the pulse width modulation control in dc-dc converter. | (3) |
| 9 | What are the various classifications of load torque. | (3) |
| 10 | With the help of a neat block diagram discuss the various parts of an electric drive. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

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|----|---|-----|
| 11 | a) With a neat circuit diagram and relevant waveforms describe briefly how firing angle control up to 180 degrees is obtained in a RC triggering circuit for SCR. | (7) |
| | b) Explain the structure and principle of operation of a power MOSFET | (7) |

OR

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| 12 | a) Describe the static and switching characteristics of a SCR. | (10) |
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- b) Deduce the Two Transistor Model for a Thyristor and briefly describe the thyristor operation. (4)

Module II

- 13 a) Draw the circuit for three phase semiconverter feeding RLE load. Explain the working with suitable output voltage waveforms for a firing angle of 60° , 120° and 150° . (10)
- b) A full wave converter feeds power to RLE load with $R=6\Omega$ and $L=6\text{mH}$ and $E=60\text{V}$. The ac source voltage is 230 V , 50 Hz . For continuous conduction, find the average load current for a firing angle of 50° . (4)

OR

- 14 a) Using suitable waveforms describe the working of a single phase full wave converter feeding to RL load for $\alpha > 90^\circ$ and $\alpha < 90^\circ$. Discuss about the power flow in each case. (10)
- b) With the help of neat waveforms derive the expression for RMS output voltage of a single-phase asymmetrical semi converter. (4)

Module III

- 15 a) Illustrate the operation of a 3-phase bridge inverter operating in 120° conduction mode with output line voltage and phase voltage waveforms (10)
- b) Give a comparison between VSI and CSI. (4)

OR

- 16 a) Illustrate the unipolar and bipolar sinusoidal pulse width modulation control in a single phase inverter. (10)
- b) Obtain an expression for THD in output voltage of a single phase full wave inverter with RL load using suitable waveforms (4)

Module IV

- 17 a) In a step down chopper the dc input voltage is of 100V . The switching frequency of MOSFET 2kHz . Find the duty cycle and average dc output voltage if the turn on period of switch is 0.2ms (7)
- b) Explain current limit control in dc-dc converters (7)

OR

- 18 a) Explain the two-quadrant operation of a dc -dc converter (7)
- b) Obtain an expression for output dc voltage in terms of input voltage and duty cycle for a buck converter with the help of necessary waveforms (7)

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Module V

- 19 a) Explain the speed control of three phase induction motor using V/f control (7)
b) Explain the simultaneous operation of a three phase dual converter. (7)

OR

- 20 a) Using suitable waveforms, describe the four quadrant operation of an ideal dual converter (7)
b) Derive the expression for output voltage of a single phase fully controlled converter drive using suitable waveforms (7)
